

Science Fact-ion

From the geek's wildest dream to reality—our fantasy gadgets are almost here!

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They say the truth is stranger than fiction. If that were so, we would already have our flying cars and servile fembots (without the machine-gun implants, please). But there are a few things that have stumbled from the world of make-believe and into our mundane lives. In the following pages we explore research work and prototype devices that have taken the first small step to bringing science fiction to fact.

Invisibility

The human dream to turn invisible is timeless. There hasn't been a medium untouched that hasn't explored this fantasy to the fascination of humans everywhere. TV shows, movies, books, videogames, radio shows, comic books—the wish to turn invisible resonates deep within each of us—be it for mischief, or otherwise.

There are several paths to invisibility—some more practical than others. In 2006, physicists in the United Kingdom suggested a mathematical model of turning object invisible. The theory was based on the principle of resonance, more specifically, resonance of light frequencies. The mathematical model suggested a cloaking material. When this material is placed next to a speck of dust that reflects light, it would resonate at the same light frequency and would in effect cancel out the ray emanating from the dust speck. This cancellation would cause the material itself to become invisible.

But light not reflecting off of an object is only half-way to true invisibility—not only should the object itself not reflect light but everything surrounding the invisible object, especially its background should be visible.

Based on that paper, Purdue researchers created a prototype device. The device consisted of an array of needles that radiated outwards from a central spoke. Based on the mathematical model of the UK researchers, this spoke would then bend light around the object that was to be cloaked. Background objects would be visible but not the object surrounded by the cylindrical array of nano-needles.

The design's limitation is that it can only cancel out a single light frequency, as it stands today. The entire gamut of visible light frequencies would need to be cancelled if this method is to be a true invisibility potion. Of course, to truly be invisible, it would also need to be not visible in the ultraviolet and the infrared frequencies... But we can all agree that even a partially invisible object is a solid step forward to fulfilling this age-old dream.

Cloaking Shield

A more mechanical form of invisibility is cloaking technology: A staple diet of aliens everywhere. In 2007, the UK army claimed to have cloaked a tank using cameras and projectors to beam images of surrounding countryside on to the surface of the tank. One can imagine this method gain traction as nanotechnology shrinks mirrors and cameras to microscopic dimensions.